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Ignition device

Abstract

For this ignition device, a fuel including hydrogen is used. The ignition device includes an ignition coil, a spark plug, a power supply device, a switching element, a first limiting diode, and a first resistor. Two first connecting wires are laid in parallel between the one end of the secondary coil and the spark plug. The first limiting diode is interposed in one first connecting wire and is forward-biased when oriented in a direction from one end to the other end of the secondary coil. The first limiting diode has a breakdown voltage that is equal to or higher than a maximum value of an ON-state voltage, and is lower than a discharge maintaining voltage of the spark plug. The first resistor is interposed in the other first connecting wire and has a resistance value of 10 MΩ or higher and of 50 MΩ or lower.

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Patent No.	Application Date	Country	CPC
6517088	12/2018	JP	N/A

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Background/Summary

RELATED APPLICATIONS

(1) This application claims the benefit of Japanese Application No. 2023-205385, filed on Dec. 5, 2023, the disclosure of which is incorporated by reference herein.

BACKGROUND OF THE INVENTION

Field of the Invention

(2) The present invention relates to an ignition device for use in an internal combustion engine.

Description of the Background Art

(3) Conventionally, an ignition device is installed in an internal combustion engine including a spark-ignition (SI) reciprocating engine used in an automobile or the like. Under control of an engine control unit (ECU), an ignition coil of the ignition device steps up a low direct-current voltage supplied from a battery to several thousands of volts to several tens of thousands of volts, and provides the stepped-up voltage to a spark plug, to generate electric spark and ignite a fuel. An example of such a conventional ignition device is described in Japanese Patent No. 6517088, for example.

(4) Japanese Patent No. 6517088 discloses an ignition device (1) for use in an internal combustion engine having the following configuration. First, a primary coil (21) of an ignition coil (2) is connected to a direct-current power supply (VB+) such as an on-vehicle battery. Then, by ON/OFF control of a main switching element (4), switching between passage and interruption of a primary current (I1) flowing through the primary coil (21) is performed (paragraph [0015] and FIG. 1).

Further, one end of a secondary coil (22) that is magnetically coupled to the primary coil (21) via an iron core is connected to a spark plug (3). The other end of the secondary coil (22) is connected to a direct-current power supply line via an ON-voltage preventing diode (23). As a result, when the primary current (I1) is interrupted in the ignition coil (2), a high voltage is generated on a secondary side, and electrical breakdown is caused at a discharge gap in the spark plug (3). Further, when the primary current (I1) is interrupted in the ignition coil (2), a secondary current (I2) flows in a forward direction of the ON-voltage preventing diode (23) (paragraphs [0016] and [0029]). Meanwhile, when the primary coil (21) starts to be energized, an ON voltage of opposite polarity generated in the secondary coil (22) is suppressed by the ON-voltage preventing diode (23) (paragraph [0017]).

(5) In recent years, a fuel including hydrogen is used in a spark-ignition (SI) reciprocating engine in many cases. It is considered that use of a fuel including hydrogen contributes to realization of a so-called low carbon society. On the other hand, however, hydrogen has properties of high combustibility at relatively low temperatures and of a high combustion rate. For this reason, for example, when a slight degree of discharge occurs in a spark plug at an unexpected timing, a fuel can possibly be ignited to burn. In this case, there is a fear of causing backfire in which flames move backward toward an intake device from a combustion chamber of an engine, after-fire in which a fuel remaining in an exhaust gas of an engine burns in an exhaust stream path or the like, or abnormal combustion such as pre-ignition in which an ignition timing is out of control.

(6) Meanwhile, in a case in which there is residual energy near a spark plug or the like at the end of discharge of the spark plug in one cycle performed in a single-cylinder or multi-cylinder combustion chamber included in an internal combustion engine, a possibility that the fuel may be ignited to burn at an unexpected timing in a next cycle is increased. More specifically, in an intake process in a next cycle, a pressure in the cylinder is reduced. Then, when a new fuel-gas mixture flows in, discharge occurs due to residual energy, which further increases a possibility that the fuel may be ignited to burn. In a case in which the number of revolutions in each cylinder is large, in particular, a next cycle is started immediately. Hence, it is necessary to cause the residual energy to converge toward zero earlier.

SUMMARY OF THE INVENTION

(7) It is an object of the present invention to provide a technology that can suppress occurrence of discharge at an unexpected timing (abnormal timing) in a spark plug. Specifically, it is an object of the present invention to provide a technology that can cause residual energy remaining near the spark plug or the like in the spark plug at the end of discharge, to converge toward zero earlier.

(8) To solve the above-described problem, the first invention of the present application is directed to an ignition device for use in an internal combustion engine that uses a fuel including at least hydrogen, and includes an ignition coil, a power supply device, a switching element, a spark plug, a first limiting diode, and a first resistor. The ignition coil is formed by electromagnetic coupling of a primary coil and a secondary coil. The power supply device applies a direct-current voltage to one end of the primary coil via a power supply line. The switching element is interposed between the other end of the primary coil and a ground point, and is capable of performing switching between passage and interruption of a primary current flowing through the primary coil from the power supply device. The spark plug ignites the fuel by occurrence of discharge at a gap, in accordance with a high voltage induced at one end of the secondary coil. The first limiting diode comprises a Zener diode or an avalanche diode that is interposed in one of two first connecting wires laid in parallel between the one end of the secondary coil and the spark plug, and is forward-biased when the diode is oriented in a direction from the one end to the other end of the secondary coil. The first resistor is interposed in the other of the two first connecting wires. The first limiting diode has a breakdown voltage that is equal to or higher than a value calculated by multiplication of a voltage value of the direct-current voltage applied to the one end of the primary coil by the power supply device by a ratio of the number of turns of the secondary coil to the number of turns of the primary

coil, and is lower than a discharge maintaining voltage at the gap in the spark plug. The first resistor has a resistance value that is equal to or higher than 10 MΩ and is equal to or lower than 50 MΩ.

(9) The second invention of the present application is directed to the ignition device of the first invention, wherein the breakdown voltage is equal to or higher than 1 kV.

(10) The third invention of the present application is directed to the ignition device of the first invention or the second invention, wherein the breakdown voltage is equal to or lower than 2 kV.

(11) The fourth invention of the present application is directed to an ignition device for use in an internal combustion engine that uses a fuel including at least hydrogen, and includes an ignition coil, a power supply device, a switching element, a spark plug, a second limiting diode, and a second resistor. The ignition coil is formed by electromagnetic coupling of a primary coil and a secondary coil. The power supply device applies a direct-current voltage to one end of the primary coil via a power supply line. The switching element is interposed between the other end of the primary coil and a ground point, and is capable of performing switching between passage and interruption of a primary current flowing through the primary coil from the power supply device. The spark plug ignites the fuel by occurrence of discharge at a gap, in accordance with a high voltage induced at one end of the secondary coil. The second limiting diode comprises a Zener diode or an avalanche diode that is interposed in one of two second connecting wires laid in parallel between the other end of the secondary coil and the power supply device or the ground point, and is forward-biased when the diode is oriented in a direction from the one end to the other end of the secondary coil. The second resistor is interposed in the other of the two second connecting wires. The second limiting diode has a breakdown voltage that is equal to or higher than a value calculated by multiplication of a voltage value of the direct-current voltage applied to the one end of the primary coil by the power supply device by a ratio of the number of turns of the secondary coil to the number of turns of the primary coil, and is lower than a discharge maintaining voltage at the gap in the spark plug. The second resistor has a resistance value that is equal to or higher than 10 MΩ and is equal to or lower than 50 MΩ.

(12) The fifth invention of the present application is directed to the ignition device of the fourth invention, wherein the breakdown voltage is equal to or higher than 1 kV.

(13) The sixth invention of the present application is directed to the ignition device of the fourth invention or the fifth invention, wherein the breakdown voltage is equal to or lower than 2 kV.

(14) The seventh invention of the present application is directed to the ignition device of any of the first to sixth inventions, which further includes a control unit configured to control the switching of the switching element. The control unit performs charge control in which the switching element is placed in a closed state, so that a primary current flows through the primary coil, to charge the primary coil, and performs discharge control in which, after the charge control, a state of the switching element is changed to an open state and a high voltage is induced at the one end of the secondary coil, so that discharge occurs at the gap in the spark plug. An absolute value of the voltage induced at the one end of the secondary coil at an ending point of the discharge control is higher than an absolute value of the voltage induced at the one end of the secondary coil at a starting point of the discharge control.

(15) The eighth invention of the present application is directed to the ignition device of any of the first to seventh inventions, which further includes a floating capacitor formed between the one end of the secondary coil and the spark plug.

(16) According to the first to eighth inventions of the present application, when a primary coil flows through the primary coil (in an ON state), a current flowing through the secondary coil can be suppressed by the limiting diode and the resistor connected in parallel with each other. This can suppress occurrence of discharge in the spark plug in an ON state. Further, after the discharge ends, an absolute value of a voltage value caused by residual energy remaining near the one end of the secondary coil, the spark plug, or the like can be significantly reduced to the breakdown voltage of

the limiting diode at once, first. Moreover, after that, a current flows via the resistor, so that the absolute value of the voltage can be reduced toward zero earlier. This can further suppress occurrence of discharge in the spark plug at an abnormal timing afterward.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a block diagram schematically showing an operating environment of an ignition device for use in an internal combustion engine according to a first preferred embodiment;
- (2) FIG. 2 is a longitudinal sectional view of an ignition coil according to the first preferred embodiment;
- (3) FIG. 3 shows a result of measuring a relationship between a resistance value of a first resistor and an ON-state voltage during flow of a primary current through a primary coil (in an ON state) according to the first preferred embodiment;
- (4) FIG. 4 is a graph showing in a time series, a waveform of an EST signal, a waveform of a current (secondary current) flowing through a secondary coil, and a waveform of a voltage (secondary voltage) applied to one end of the secondary coil at the time of activating the ignition device according to the first preferred embodiment;
- (5) FIG. 5 shows a result of measuring, by simulation, a time taken for an absolute value of a voltage (secondary voltage) applied to one end of the secondary coil to converge to 200 V while changing an initial residual voltage, with the use of the ignition device according to the first preferred embodiment;
- (6) FIG. 6 is a block diagram schematically showing an operating environment of an ignition device for use in an internal combustion engine according to a second preferred embodiment;
- (7) FIG. 7 is a block diagram schematically showing an operating environment of an ignition device for use in an internal combustion engine according to a first modification; and
- (8) FIG. 8 is a block diagram schematically showing an operating environment of an ignition device for use in an internal combustion engine according to a second modification.

DESCRIPTION OF THE PREFERRED EMBODIMENTS

(9) Below, illustrative preferred embodiments of the present invention will be described with reference to the drawings. Note that components described in these embodiments are merely examples and are not intended to limit the scope of the present invention to those only. Further, in the drawings, for the purpose of easier understanding, the dimensions or the number of respective components are overstated or understated in some portions of illustration, as necessary.

1. First Preferred Embodiment

1-1. Configuration of Ignition Device

(10) First, a configuration of an ignition device **1** for use in an internal combustion engine corresponding to a first preferred embodiment of the present invention will be described with reference to the drawings. FIG. 1 is a block diagram schematically showing an operating environment of the ignition device **1** according to the first preferred embodiment. Note that a primary coil **L1** and a secondary coil **L2** of an ignition coil **103** included in the ignition device **1** are arranged so as to be stacked on each other as described later. However, in FIG. 1, the primary coil **L1** and the secondary coil **L2** are shown as being arranged adjacent to each other, for the purpose of easy understanding.

(11) The ignition device **1** according to the present embodiment is, for example, a device that is installed in an internal combustion engine such as a spark-ignition (SI) reciprocating engine used in a vehicle body **100** of an automobile or the like and applies a high voltage for causing spark discharge to occur in a spark plug **113**. The ignition device **1** is provided in a cylinder or each of a plurality of cylinders included in the internal combustion engine.

(12) Further, as shown in FIG. 1, the vehicle body **100** is equipped with the spark plug **113**, a power supply device **102** (battery), and an engine control unit (ECU) **105**, in addition to the ignition device **1**. Note that, in a broad sense, the spark plug **113**, the power supply device **102**, and the ECU **105** can be regarded as being included in the ignition device **1**.

(13) The spark plug **113** is a device for performing an ignition operation in a combustion chamber of the internal combustion engine. The spark plug **113** is electrically connected to one end **822** of the secondary coil L2 of the ignition coil **103** described later via a conductor. Hereinafter, this conductor will be referred to as a “first connecting wire **121**”. The spark plug **113** is interposed between the one end **822** of the secondary coil L2 and a ground point (ground) **151**. A high voltage is induced in the secondary coil L2 of the ignition coil **103**. Then, when the high voltage exceeds an electrical breakdown voltage at a gap *d* (refer to FIG. 1) between a center electrode **161** and a ground electrode **162** of the spark plug **113**, discharge occurs at the gap *d*, so that spark is generated. As a result, a fuel supplied to the internal combustion engine is ignited. In other words, the spark plug **113** ignites a fuel by occurrence of discharge at the gap *d*, in accordance with a high voltage induced at the one end **822** of the secondary coil L2.

(14) In the present embodiment, hydrogen or a mixture of hydrogen and other materials is used as a fuel. That is, a fuel including at least hydrogen is used in the ignition device **1** for use in an internal combustion engine.

(15) Further, in the first connecting wire **121** and the spark plug **113**, an electrostatic capacitance component of approximately 15 to 20 pF is present. In other words, an electrostatic capacitance component is formed between the one end **822** of the secondary coil L2 and the spark plug **113**. Hereinafter, the electrostatic capacitance component will be referred to as a “floating capacitor *C_s*” that is virtually defined. As shown in FIG. 1, the floating capacitor *C_s* can be schematically expressed in parallel with the spark plug **113** in the block diagram.

(16) The power supply device **102** is a direct-current power chargeable/dischargeable device (storage battery). In the present embodiment, the power supply device **102** is electrically connected to the primary coil L1 of the ignition coil **103** described later via a conductor. Hereinafter, this conductor will be referred to as a “power supply line **150**”. The power supply device **102** applies a direct-current voltage to one end **811** of the primary coil L1 of the ignition coil **103** via the power supply line **150**.

(17) The ECU **105** is an existing computer that comprehensively controls operations and the like of a transmission and an air bag in the vehicle body **100**.

(18) The ignition device **1** includes the ignition coil **103**, an igniter **104**, a first limiting diode **131**, and a first resistor **132**.

(19) FIG. 2 is a longitudinal sectional view of the ignition coil **103**. As shown in FIG. 2, the ignition coil **103** includes a bobbin **40**, the primary coil L1, the secondary coil L2, and an iron core **60**. Note that parts of the primary coil L1 and the secondary coil L2 are shown in a simplified manner in FIG. 2. Meanwhile, in the following description about the ignition coil **103**, a direction parallel with a center axis Bc of the bobbin **40** will be referred to as an “axis direction”. A direction perpendicular to the center axis Bc of the bobbin **40** will be referred to as a “diameter direction”. A direction along an arc having its center on the center axis Bc of the bobbin **40** will be referred to as a “circumference direction”. Further, the “direction parallel with something” includes a direction substantially parallel with something, and the “direction perpendicular to something” includes a direction substantially perpendicular to something.

(20) The bobbin **40** includes a primary bobbin **41** and a secondary bobbin **42** that can be coupled to each other. Each of the primary bobbin **41** and the secondary bobbin **42** extends in a tubular shape along the center axis Bc. Further, the secondary bobbin **42** is placed on the diameter-direction outer side of the primary bobbin **41**. For a material of the primary bobbin **41** and the secondary bobbin **42**, resin is used, for example.

(21) The primary coil L1 is formed by winding of a conductor around an outer surface of the

primary bobbin **41** in the circumference direction having its center on the center axis Bc. Hereinafter, this conductor will be referred to as a “primary conductor **81**”. After the primary coil L1 is formed, the secondary bobbin **42** is placed so as to cover the outer surface of the primary coil L1, and is coupled to the primary bobbin **41**. Then, a conductor different from the primary conductor **81** is wound around the outer surface of the secondary bobbin **42** in the circumference direction having its center on the center axis Bc, to thereby form the secondary coil L2. Hereinafter, this different conductor will be referred to as a “secondary conductor **82**”. By arranging the primary coil L1 and the secondary coil L2 such that the coils are stacked on each other in the above-described manner, it is possible to miniaturize the entire ignition coil **103** including those coils. However, arrangement of the primary coil L1 and the secondary coil L2 is not limited to the above-described case in which the coils are stacked on each other with the conductors being wound therearound. Alternatively, the primary coil L1 and the secondary coil L2 may be arranged adjacent to each other as shown in FIG. 1.

(22) The iron core **60** has a structure in which a central iron core **601** and an outer iron core **602** are combined. Each of the central iron core **601** and the outer iron core **602** of the iron core **60** is formed of a laminated steel sheet in which silicon steel sheets are stucked together, for example. The central iron core **601** extends along the center axis Bc of the bobbin **40**. Further, the central iron core **601** is inserted through a space **410** on the diameter-direction inner side of the primary bobbin **41**. The outer iron core **602** extends on the diameter-direction outer side with respect to the secondary bobbin **42** and the secondary conductor **82**, and connects both axial ends of the central iron core **601**. Thus, the iron core **60** forms a closed magnetic circuit structure that electromagnetically couples the primary coil L1 and the secondary coil L2. In other words, the ignition coil **103** is formed by electromagnetic coupling of the primary coil L1 and the secondary coil L2.

(23) As shown in FIG. 1, the one end **811** of the primary coil L1 is connected to the power supply line **150** that is the above-described conductor extending from the power supply device **102**. The other end **812** of the primary coil L1 is connected to the igniter **104** described later. Under control of the igniter **104**, a low direct-current voltage supplied from the power supply device **102** is applied to the one end **811** of the primary coil L1, and then a primary current that gradually increases starts to flow through the primary coil L1.

(24) The one end **822** of the secondary coil L2 is connected to the spark plug **113** via the first connecting wire **121**. The secondary conductor **82** has a wire size smaller than a wire size of the primary conductor **81**. Further, the number of turns (for example, 8000) of the secondary conductor **82** in the secondary coil L2 is approximately 80 times or more the number of turns (for example, 100) of the primary conductor **81** in the primary coil L1. Thus, the ignition coil **103** steps up power of a low direct-current voltage supplied from the power supply device **102** to several thousands of volts to several tens of thousands of volts during interruption of a primary current, details of which will be given later. That is, a high voltage is induced in the secondary coil L2. Then, the secondary coil L2 supplies power of the induced high voltage to the spark plug **113**. Consequently, electric spark is generated in the spark plug **113**, and a fuel is ignited.

(25) Meanwhile, the first connecting wire **121** includes two conductors arranged in parallel with each other. Hereinafter, these two conductors will be referred to as a “first connecting wire **121a**”, and a “first connecting wire **121b**”. Specifically, the two first connecting wires **121a** and **121b** are laid in parallel with each other between the one end **822** of the secondary coil L2 and the spark plug **113**.

(26) Further, in the present embodiment, in the first connecting wire **121a** that is one of the two first connecting wires **121a** and **121b**, the first limiting diode **131** is interposed. The first limiting diode **131** is connected in series to the secondary coil L2. For the first limiting diode **131** of the present embodiment, a Zener diode is used. Alternatively, for the first limiting diode **131**, an avalanche diode may be used. Moreover, the first limiting diode **131** is forward-biased when it is

oriented in a direction from the one end **822** to the other end **821** of the secondary coil **L2**.

(27) Further, in the present invention, for the first limiting diode **131**, a diode having a breakdown voltage that is equal to or higher than a maximum value of an “ON-state voltage” described later and is lower than a “discharge maintaining voltage” at the gap **d** in the spark plug **113**, is used. Note that the terms, “discharge maintaining voltage”, mean a voltage to be applied to the gap **d** between the center electrode **161** and the ground electrode **162** of the spark plug **113** in order to maintain discharge at the gap **d**. The first limiting diode **131** of the present embodiment has a breakdown voltage that is equal to or higher than 1 kV and is equal to or lower than 2 kV. An effect produced by setting the breakdown voltage of the first limiting diode **131** to the above-mentioned value will be later described in detail.

(28) Further, in the present embodiment, in the first connecting wire **121b** that is the other of the two first connecting wires **121a** and **121b**, the first resistor **132** is interposed. The first resistor **132** is connected in series to the secondary coil **L2**. Moreover, the first resistor **132** of the present embodiment has a resistance value that is equal to or higher than 10 MΩ and is equal to or lower than 50 MΩ. An effect produced by setting the resistance value of the first resistor **132** to the above-mentioned value will be later described in detail.

(29) In addition, as shown in FIG. **1**, in the secondary coil **L2**, the other end **821** opposite to the one end **822** connected to the spark plug **113** is electrically connected directly or indirectly to the power supply device **102** via a conductor. Hereinafter, this conductor will be referred to as a “second connecting wire **122**”. In the present embodiment, the other end **821** of the secondary coil **L2** is electrically connected to the power supply line **150** via the second connecting wire **122**.

(30) When a switching element **70** of the igniter **104** is placed in a closed state and a primary current flows through the primary coil **L1** to charge the primary coil **L1**, a potential difference is caused between the one end **822** and the other end **821** of the secondary coil **L2**, details of which will be given later. Hereinafter, this period in which a primary current flows through the primary coil **L1** to charge the primary coil **L1** will be referred to as an “ON state”. It depends on the winding direction of the secondary coil **L2** whether a voltage to be induced across the one end **822** and the other end **821** of the secondary coil **L2** is positive or negative. In the present embodiment, in an ON state, the one end **822** of the secondary coil **L2** is at a voltage level higher than the other end **821**. Meanwhile, hereinafter, a potential difference between the one end **822** and the other end **821** of the secondary coil **L2** will be referred to as an “ON-state voltage”. A maximum value of the ON-state voltage is calculated by multiplication of a voltage value of a direct-current voltage applied to the one end **811** of the primary coil **L1** by the power supply device **102** via the power supply line **150**, by a ratio of the number of turns of the secondary coil **L2** to the number of turns of the primary coil **L1**.

(31) For example, suppose a case in which a voltage value of a direct-current voltage applied to the one end **811** of the primary coil **L1** is 12 V, the number of turns of the primary coil **L1** is 100, and the number of turns of the secondary coil **L2** is 8000. Then, a ratio of the number of turns of the secondary coil **L2** to the number of turns of the primary coil **L1** is 80, and hence a maximum value of the ON-state voltage is calculated as $12 \times 80 = 960$ V. Thus, a maximum value of a voltage applied to the one end **822** of the secondary coil **L2** is, for example, approximately plus 480 V, and a minimum value of a voltage applied to the other end **821** of the secondary coil **L2** is, for example, approximately minus 480 V. Further, depending on the circumstances, it can be supposed that a maximum value of a voltage applied to the one end **822** of the secondary coil **L2** is approximately 0 V, and a minimum value of a voltage applied to the other end **821** of the secondary coil **L2** is approximately minus 960 V. Meanwhile, at that time, a voltage applied to the power supply line **150** is 12 V.

(32) Thus, a current directed toward the secondary coil **L2** flows from the side where the power supply device **102** is present, via the power supply line **150** and the second connecting wire **122**. It should be noted here that the first limiting diode **131** is interposed in the first connecting wire **121a**

of the first connecting wire **121** connecting the one end **822** of the secondary coil **L2** and the spark plug **113**. As described above, the breakdown voltage of the first limiting diode **131** is equal to or higher than 1 kV. That is, in the present embodiment, the breakdown voltage of the first limiting diode **131** is set so as not to fall below a maximum value of a voltage (differential voltage) applied to the one end **822** of the secondary coil **L2** with respect to the ground point (ground) **151** during flow of a primary current through the primary coil **L1**. In other words, the breakdown voltage of the first limiting diode **131** is set so as not to fall below a maximum value of a voltage (differential voltage) applied to a cathode side of the first limiting diode **131** with respect to a voltage applied to an anode side of the first limiting diode **131** in an ON state. Hence, no current flows through the first connecting wire **121** in which the first limiting diode **131** is interposed.

(33) Further, in the first connecting wire **121b** of the first connecting wire **121**, the first resistor **132** is interposed. As described above, the resistance value of the first resistor **132** is equal to or higher than 10 MΩ. By setting the resistance value of the first resistor **132** to such a sufficiently high value as described above, it is possible to sufficiently reduce a current flowing via the first connecting wire **121b** in an ON state. As a result of this, in an ON state, an ON-state voltage generated in the secondary coil **L2** and a voltage (secondary voltage) generated at the one end **822** of the secondary coil **L2** can be suppressed. This can suppress occurrence of discharge in the spark plug **113** in an ON state, that is, at an abnormal timing. Meanwhile, for reference, a result of measuring a relationship between the resistance value of the first resistor **132** and an ON-state voltage during flow of a primary current through the primary coil **L1** (in an ON state) with the use of the ignition device **1** according to the present embodiment, is shown in FIG. 3. Note that FIG. 3 shows a result of measuring a relationship between the resistance value of the first resistor **132** and an ON-state voltage while changing the resistance value of the first resistor **132** to various values.

(34) The igniter **104** is a semiconductor device that is connected to the primary coil **L1** and controls a current flowing through the primary coil **L1**. Further, the igniter **104** is electrically connected to the ECU **105** and receives a signal from the ECU **105**. Hereinafter, this signal will be referred to as an “EST signal”. The igniter **104** includes the switching element **70** and a drive IC **71**. The igniter **104** may be integral with an electronic circuit of the ECU **105**.

(35) For the switching element **70**, an insulated-gate bipolar transistor (IGBT) is used, for example. The switching element **70** is interposed between the other end **812** of the primary coil **L1** and a ground point (ground) **152**. A collector (C) of the switching element **70** is connected to the other end **812** of the primary coil **L1**. An emitter (E) of the switching element **70** is connected to the ground point (ground) **152**. A gate (G) of the switching element **70** is connected to the drive IC **71**.

(36) Thus, the switching element **70** can perform switching between passage and interruption of a primary current flowing through the primary coil **L1** from the power supply device **102**. When the switching element **70** is placed in a closed state, a primary current flows through the primary coil **L1** from the power supply device **102**. When the switching element **70** is placed in an open state, a primary current flowing through the primary coil **L1** is interrupted. Meanwhile, other kinds of transistors may be used for the switching element **70**.

(37) The drive IC **71** is a control unit that controls switching of the switching element **70** in accordance with an EST signal received from the ECU **105**. The drive IC **71** includes a logic device connected to the switching element **70**. The logic device includes, for example, a logic circuit, a processor, a complex programmable logic device (CPLD), a field-programmable gate array (FPGA), an application-specific integrated circuit (ASIC), or the like. The logic device performs arithmetic processing for activating the ignition device **1**, to achieve ignition in the spark plug **113**.

1-2. Operations of Ignition Device

(38) Next, operations of the ignition device **1** will be described. FIG. 4 shows graphs respectively representing in a time series, a waveform of an EST signal, a waveform of a current (secondary current) flowing through the secondary coil **L2**, and a waveform of a voltage (secondary voltage) generated at the one end **822** of the secondary coil **L2** at the time of activating the ignition device **1**.

Note that, regarding a secondary current in FIG. 4, “NEGATIVE” represents a direction from the one end **822** to the other end **821** of the secondary coil L2, and “POSITIVE” represents a direction from the other end **821** to the one end **822** of the secondary coil L2. Further, regarding a secondary voltage in FIG. 4, a value of a voltage applied to the one end **822** of the secondary coil L2 with respect to the ground point (ground) is shown.

(39) As described above, a direct-current voltage is applied to the one end **811** of the primary coil L1 by the power supply device **102** via the power supply line **150**. For example, a direct-current voltage of 12 V is applied to the one end **811** of the primary coil L1. Meanwhile, the other end **812** of the primary coil L1 is connected to the switching element **70**. Further, the drive IC **71** controls switching of the switching element **70** in accordance with an EST signal received from the ECU **105**. Note that, in the present embodiment, a fuel-gas mixture including a fuel is supplied to the combustion chamber of the single cylinder or each of the plurality of cylinders of the internal combustion engine so that the combustion chamber is filled with the fuel, and then, each ignition device **1** is activated immediately before a top dead center (TDC). As shown in FIG. 4, in activating the ignition device **1**, first, the signal level of an EST signal transmitted from the ECU **105** to the drive IC **71** is changed from L to H at a time **t0**.

(40) Then, the drive IC **71** changes the state of the switching element **70** from an open state to a closed state in accordance with the EST signal. This causes a primary current to flow through the primary conductor **81** forming the primary coil L1, so that the primary coil L1 is charged with electric charge. Hereinafter, such a process in which a primary current flows through the primary coil L1 to charge the primary coil L1 will be referred to as “charge control”. Further, an energization magnetic flux is generated in the primary coil L1, and a magnetic field corresponding to the energization magnetic flux acts on the iron core **60**.

(41) Moreover, a potential difference, that is, an ON-state voltage, is generated across both ends **821** and **822** of the secondary coil L2 electromagnetically coupled to the primary coil L1 via the iron core **60** by an effect of mutual induction. For example, a potential difference of 960 V is generated across the one end **822** and the other end **821**. As a result, a maximum value of a voltage applied to the one end **822** of the secondary coil L2 is positive, and a minimum value of a voltage applied to the other end **821** of the secondary coil L2 is negative. For example, a maximum value of a voltage applied to the one end **822** of the secondary coil L2 is approximately plus 480 V, and a minimum value of a voltage applied to the other end **821** of the secondary coil L2 is approximately minus 480 V. At that time, a voltage applied to the power supply line **150** is, for example, 12 V.

(42) Thus, a current directed toward the secondary coil L2 flows from the side where the power supply device **102** is present, via the power supply line **150** and the second connecting wire **122**. It should be noted here that the first limiting diode **131** is interposed in the first connecting wire **121a** of the first connecting wire **121** connecting the one end **822** of the secondary coil L2 and the spark plug **113**. As described above, the breakdown voltage of the first limiting diode **131** is equal to or higher than 1 kV. That is, in the present embodiment, the breakdown voltage of the first limiting diode **131** is set to a value that does not fall below a maximum value of a voltage (differential voltage) applied to the one end **822** of the secondary coil L2 with respect to the ground point (ground) **151** during flow of a primary current through the primary coil L1. In other words, the breakdown voltage of the first limiting diode **131** is set to a value that does not fall below a maximum value of a voltage (differential voltage) applied to the cathode side of the first limiting diode **131** with respect to a voltage applied to the anode side of the first limiting diode **131** in an ON state. For this reason, no current flows through the first connecting wire **121a** in which the first limiting diode **131** is interposed.

(43) Further, the first resistor **132** is interposed in the first connecting wire **121b** of the first connecting wire **121**. As described above, the resistance value of the first resistor **132** is equal to or higher than 10 MΩ. By setting the resistance value of the first resistor **132** to such a sufficiently high value as described, it is possible to sufficiently reduce a current flowing via the first

connecting wire **121b** in an ON state. As a result of this, in an ON state, an ON-state voltage generated in the secondary coil **L2** or a voltage (secondary voltage) generated at the one end **822** of the secondary coil **L2** can be suppressed. Consequently, occurrence of discharge in the spark plug **113** in an ON state, that is, at an abnormal timing, can be suppressed.

(44) After the charge control, the signal level of the EST signal transmitted from the ECU **105** to the drive IC **71** is changed from H to L at a time **t1**. Then, the drive IC **71** changes the state of the switching element **70** from a closed state to an open state, to interrupt a primary current flowing from the power supply device **102** to the primary coil **L1**. As a result, induced electromotive force is generated in the secondary coil **L2** electromagnetically coupled to the primary coil **L1** via the iron core **60** by an effect of mutual induction. In the present embodiment, a high negative voltage is induced at the one end **822** of the secondary coil **L2**. At that time, a value of a voltage applied to the one end **822** of the secondary coil **L2** (i.e., a value of a secondary voltage) ranges from minus several thousands of volts to minus several tens of thousands of volts with respect to the ground point (ground).

(45) Further, an absolute value of the high negative voltage induced at the one end **822** of the secondary coil **L2** exceeds the electrical breakdown voltage at the gap **d** in the spark plug **113**. This causes electrical breakdown at the gap **d** in the spark plug **113**. Then, there is generated a current that flows from the ground point (ground) **151** to the center electrode **161** of the spark plug **113** via the ground electrode **162** of the spark plug **113** (refer to FIG. 1). The thus generated current flows through the first limiting diode **131** in the forward direction, or flows through the first resistor **132**, and further flows through the secondary coil **L2**. In the present embodiment, most of the current flows through the first limiting diode **131** in the forward direction while a part thereof flows through the first resistor **132**, and the current further flows to a ground point (ground) **153** via the power supply device **102**.

(46) As a result, discharge occurs at the gap **d** in the spark plug **113**, so that spark is generated and a fuel supplied to the combustion chamber of the internal combustion engine is ignited. When the fuel in the combustion chamber burns, the pressure in the combustion chamber increases, so that a piston moves from a top dead center (TDC) to a bottom dead center (BDC).

(47) Note that, in the present invention, such a process in which the state of the switching element **70** is changed to an open state, a primary current flowing through the primary coil **L1** is interrupted, and a high voltage is induced at the one end **822** of the secondary coil **L2**, to cause discharge at the gap **d** in the spark plug **113** will be referred to as “discharge control”. In addition, when an absolute value of the high negative voltage induced at the one end **822** of the secondary coil **L2** falls below the discharge maintaining voltage at the gap **d** in the spark plug **113** (at a time **t2**), the discharge at the gap **d** in the spark plug **113** ends once.

(48) In this regard, the electrical breakdown voltage and the discharge maintaining voltage at the gap **d** in the spark plug **113** are significantly affected by the pressure in the combustion chamber. A voltage for causing electrical breakdown at the gap **d** in the spark plug **113** and a voltage for maintaining discharge at the gap **d** are substantially proportional to the pressure in the combustion chamber. Note that these voltages represent each an absolute value of the high voltage induced at the one end **822** of the secondary coil **L2**. Meanwhile, by the “discharge control”, the fuel in the combustion chamber burns, so that the pressure increases, as described above. Thus, in a case in which the pressure in the combustion chamber becomes high, or the fuel burns and vigorously flows, the absolute value of the high voltage induced at the one end **822** of the secondary coil **L2** becomes higher in some cases.

(49) As shown in FIG. 4, also in the present embodiment, by the “discharge control”, the absolute value of the high negative voltage induced at the one end **822** of the secondary coil **L2** temporarily approaches zero from a value (**Ds**) at a starting point of the “discharge control”, but increases again due to subsequent increase of the pressure in the combustion chamber or vigorous flow of the fuel. Then, a value (**De**) at an ending point of the “discharge control” has an absolute value larger than

that of the value (Ds) at the starting point of the “discharge control”. That is, in the present embodiment, the absolute value of the voltage induced at the one end **822** of the secondary coil **L2** at the ending point of the discharge control is larger than the absolute value of the voltage induced at the one end **822** of the secondary coil **L2** at the starting point of the discharge control.

(50) Further, as described above, the floating capacitor Cs comprising an electrostatic capacitance component of approximately 15 to 20 pF is formed between the one end **822** of the secondary coil **L2** and the spark plug **113**. Because of this, in some cases, electric charge still remains in spots such as a spot near the one end **822** of the secondary coil **L2**, the first connecting wire **121**, or a spot near the center electrode **161** of the spark plug **113** even at a time (the time **t2**) when the discharge at the gap d in the spark plug **113** ends once. In the present embodiment, negative electric charge remains in those spots. As a result, a residual voltage value at the one end **822** of the secondary coil **L2** is negative with respect to the ground point (ground) at the time **t2**. Note that the residual voltage value represents the value (De) at the ending point of the “discharge control”. At the time **t2**, the residual voltage value at the one end **822** of the secondary coil **L2** is, for example, minus several tens of thousands of volts with respect to the ground point (ground).

(51) Meanwhile, at the time **t2**, the pressure in the combustion chamber is high, and the absolute value of the above-described residual voltage value is lower than the discharge maintaining voltage at the gap d in the spark plug **113**. However, to leave this condition unattended might possibly cause discharge to occur again at the gap d in the spark plug **113** at an unexpected timing such as a time when the pressure in the internal combustion engine is changed afterward. Note that, as an unexpected timing, a time when the pressure in the cylinder is reduced and a new fuel-gas mixture flows in during an intake process or the like in a next cycle, is cited, for example.

(52) Then, according to the present invention, a diode having a breakdown voltage that is lower than the discharge maintaining voltage at the gap d in the spark plug **113** and the absolute value of the above-described residual voltage value (De) is used as the first limiting diode **131**. The breakdown voltage of the first limiting diode **131** used in the present embodiment is equal to or lower than 2 kV. Meanwhile, in the above-described example, the residual voltage value (De) at the one end **822** of the secondary coil **L2** is negative, and the absolute value of the corresponding numerical value exceeds the breakdown voltage of the first limiting diode **131**. Note that the residual voltage value (De) at the one end **822** of the secondary coil **L2** is the residual voltage value (De) on the cathode side of the first limiting diode **131** and is, for example, minus several tens of thousands of volts.

(53) This allows a current to flow suddenly and abundantly through the first limiting diode **131** in the reverse direction from the side where the power supply device **102** is present, without occurrence of further discharge in the spark plug **113** after the end of discharge. Specifically, a current (secondary current) flows toward the secondary coil **L2** from the side where the power supply device **102** is present, via the second connecting wire **122**.

(54) Consequently, electric charge remaining in spots such as a spot near the one end **822** of the secondary coil **L2**, the first connecting wire **121**, or a spot near the center electrode **161** of the spark plug **113** is removed, and thus the absolute value of the voltage (secondary voltage) applied to the one end **822** of the secondary coil **L2** is significantly reduced at once, so that residual energy remaining in those spots can be reduced. Thus, the absolute value of the voltage applied to the center electrode **161** of the spark plug **113** is significantly reduced to the breakdown voltage of the first limiting diode **131** at once. Hereinafter, the thus reduced absolute value of the voltage (secondary voltage) applied to the one end **822** of the secondary coil **L2** will be referred to as an “initial residual voltage Vo”. As shown in FIG. 4, at a time (time **t2**) when the discharge ends, the absolute value of the voltage (secondary voltage) applied to the one end **822** of the secondary coil **L2** is significantly reduced at once, and the value after being reduced can be regarded as being the “initial residual voltage Vo”.

(55) Further, as described above, the first connecting wire **121b** is laid in parallel with the first

connecting wire **121a** in which the first limiting diode **131** is interposed, and the first resistor **132** is interposed in the first connecting wire **121b**. After the absolute value of the voltage applied to the center electrode **161** of the spark plug **113** is reduced to the breakdown voltage (initial residual voltage V_0) of the first limiting diode **131**, a current flows mainly via the first connecting wire **121b**. More specifically, a current (secondary current) flows through the first connecting wire **121b** in which the first resistor **132** is interposed, from the side where the power supply device **102** is present, via the second connecting wire **122** and the secondary coil **L2**.

(56) Further, in the present embodiment, the resistance value of the first resistor **132** is equal to or lower than 50 MΩ. By setting the resistance value of the first resistor **132** to such a low value as described, it is possible to keep a certain amount of current or more flowing from the side where the power supply device **102** is present, via the first connecting wire **121b** after the end of discharge. This enables further reduction of residual energy remaining near the one end **822** of the secondary coil **L2**, in the first connecting wire **121**, near the center electrode **161** of the spark plug **113**, or the like, and allows the energy to converge toward zero earlier.

(57) Meanwhile, in the ignition device **1** according to the present embodiment, an absolute value “ V_t ” of a voltage after elapse of a time period t from the time when the absolute value of the voltage (secondary voltage) applied to the one end **822** of the secondary coil **L2** becomes equal to the breakdown voltage (initial residual voltage V_0) of the first limiting diode **131** can be calculated by an expression of “ $V_t = V_0 \times \exp(-t/(C \times R))$ ”. Note that, in the expression, “ C ” represents a value of the above-described “floating capacitor C_s ”, and “ R ” represents the resistance value of the first resistor **132**. Further, the “voltage after elapse of a time period t ” represents the “voltage that has been reduced”.

(58) In addition, for reference, FIG. 5 shows a result of measuring, by simulation, a time period taken for the absolute value of the voltage (secondary voltage) applied to the one end **822** of the secondary coil **L2** to converge to 200 V, with the use of the ignition device **1** according to the present embodiment in each of respective cases in which the resistance value of the first resistor **132** is “10 MΩ”, “22 MΩ”, and “33 MΩ”. Note that FIG. 5 shows a result of measuring a time period taken for the absolute value of the voltage applied to the one end **822** of the secondary coil **L2** to converge to 200 V while changing the initial residual voltage V_0 [kV].

(59) As described above, in the present embodiment, the “initial residual voltage V_0 ” is significantly reduced to the breakdown voltage of the first limiting diode **131** at once at the time (time t_2) when the discharge ends. Specifically, the “initial residual voltage V_0 ” is significantly reduced to 2 kV or lower at once. Thus, as shown in FIG. 5, it has been confirmed that the absolute value of the voltage (secondary voltage) applied to the one end **822** of the secondary coil **L2** converges to 200 V within approximately 3.5 milliseconds even in the cases in which the resistance value of the first resistor **132** is “10 MΩ”, “22 MΩ”, and “33 MΩ”.

(60) It should be noted here that the ignition device **1** of the present embodiment is used for a high-speed internal combustion engine as described above. For example, in a case in which discharge control is performed near a top dead center (TDC) in an internal combustion engine of which number of revolutions is 16,000 rpm, an intake process in a next cycle is started approximately 3.75 milliseconds later. As described above, in the present embodiment, even in the cases in which the resistance value of the first resistor **132** is “10 MΩ”, “22 MΩ”, and “33 MΩ”, the absolute value of the voltage (secondary voltage) applied to the one end **822** of the secondary coil **L2** converges to 200 V within approximately 3.5 milliseconds. Thus, it has been confirmed that a fuel is prevented from being ignited due to occurrence of discharge also in an intake process or the like in a next cycle.

(61) As is made clear from the foregoing description, in the present embodiment, first, for charge control, when a primary current flows through the primary coil **L1** (in an ON state), a current flowing through the secondary coil **L2** can be suppressed by the first limiting diode **131** and the first resistor **132** connected in parallel. This enables reduction of an ON-state voltage generated in

the secondary coil L2. Consequently, occurrence of discharge in the spark plug **113** in an ON state can be suppressed.

(62) Further, after the end of discharge, a current (secondary current) flows suddenly and abundantly through the first limiting diode **131** in the reverse direction from the side where the power supply device **102** is present, via the second connecting wire **122** and the secondary coil L2. Thus, electric charge remaining in spots such as a spot near the one end **822** of the secondary coil L2, the first connecting wire **121**, or a spot near the center electrode **161** of the spark plug **113** can be removed at once. As a result, the absolute value of the voltage value caused by residual energy remaining in those spots can be significantly reduced to the breakdown voltage of the first limiting diode **131** at once, first. Further, after that, a current flows via the first resistor **132**, so that the absolute value of the voltage value caused by the residual energy remaining in those spots can be reduced toward zero earlier.

(63) Consequently, even in a case in which there is a change in the pressure in the internal combustion engine afterward, occurrence of discharge at the gap d in the spark plug **113** at an abnormal timing can be suppressed. This prevents a fuel from being ignited at an abnormal timing also in an internal combustion engine that uses a fuel including hydrogen having properties of high combustibility at relatively low temperatures and of a high combustion rate, leading to suppression of breakage of the engine and the like.

2. Second Preferred Embodiment

(64) Next, a second preferred embodiment of the present invention will be described. Note that, in the following description, differences from the first preferred embodiment will be mainly described, and duplicated description of parts similar to the first preferred embodiment will be omitted.

(65) FIG. **6** is a block diagram schematically showing an operating environment of the ignition device **1** according to the second preferred embodiment. As shown in FIG. **6**, in the second preferred embodiment, the one end **822** of the secondary coil L2 is electrically connected directly or indirectly to the spark plug **113** via a conductor. Hereinafter, this conductor will be referred to as a “first connecting wire **221**”.

(66) Further, in the secondary coil L2, the other end **821** opposite to the one end **822** connected to the spark plug **113** is electrically connected directly or indirectly to the power supply device **102** via two conductors. Hereinafter, these two conductors will be referred to as a “second connecting wire **222a**” and a “second connecting wire **222b**”. The second connecting wires **222a** and **222b** are laid in parallel between the other end **821** of the secondary coil L2 and the power supply device **102**. In the present embodiment, the other end **821** of the secondary coil L2 is electrically connected to the power supply line **150** via the second connecting wire **222a** or the second connecting wire **222b**.

(67) Further, in the present embodiment, in the second connecting wire **222a** that is one of the two second connecting wires **222a** and **222b**, a second limiting diode **231** is interposed. The second limiting diode **231** is connected in series to the secondary coil L2. For the second limiting diode **231** of the present embodiment, a Zener diode is used. Alternatively, an avalanche diode may be used for the second limiting diode **231**. Moreover, the second limiting diode **231** is forward-biased when it is oriented in a direction from the one end **822** to the other end **821** of the secondary coil L2.

(68) Further, in the present embodiment, for the second limiting diode **231**, a diode having a breakdown voltage that is equal to or higher than a maximum value of an ON-state voltage and is lower than the discharge maintaining voltage at the gap d in the spark plug **113**, is used. Specifically, the breakdown voltage of the second limiting diode **231** is equal to or higher than a value calculated by multiplication of a voltage value of a direct-current voltage applied to the one end **811** of the primary coil L1 by the power supply device **102** via the power supply line **150** by a ratio of the number of turns of the secondary coil L2 to the number of turns of the primary coil L1.

More specifically, the breakdown voltage of the second limiting diode **231** of the present embodiment is equal to or higher than 1 kV and is equal to or lower than 2 kV.

(69) Further, in the present embodiment, in the second connecting wire **222b** that is the other of the two second connecting wires **222a** and **222b**, a second resistor **232** is interposed. The second resistor **232** is connected in series to the secondary coil **L2**. The second resistor **232** of the present embodiment has a resistance value that is equal to or higher than 10 MΩ and is equal to or lower than 50 MΩ. Moreover, the floating capacitor **Cs** comprising an electrostatic capacitance component of approximately 15 to 20 pF is formed between the one end **822** of the secondary coil **L2** and the spark plug **113**.

(70) In the present embodiment, first, for charge control, when a primary current flows through the primary coil **L1** (in an ON state), an ON-state voltage is generated across both ends **821** and **822** of the secondary coil **L2**. For example, a potential difference of 960 V is generated across the one end **822** and the other end **821** of the secondary coil **L2**. A maximum value of a voltage applied to the one end **822** of the secondary coil **L2** is positive, and a minimum value of a voltage applied to the other end **821** of the secondary coil **L2** is negative. For example, a maximum value of a voltage applied to the one end **822** of the secondary coil **L2** is approximately plus 480 V, and a minimum value of a voltage applied to the other end **821** of the secondary coil **L2** is approximately minus 480 V. Meanwhile, at that time, a voltage applied to the power supply line **150** is, for example, 12 V.

(71) Thus, a current directed toward the secondary coil **L2** flows from the side where the power supply device **102** is present, via the power supply line **150**. It should be noted here that the second limiting diode **231** is interposed in the second connecting wire **222a** connecting the other end **821** of the secondary coil **L2** and the power supply line **150**. As described above, the second limiting diode **231** is forward-biased when it is oriented in a direction from the one end **822** to the other end **821** of the secondary coil **L2**. Further, the breakdown voltage of the second limiting diode **231** is equal to or higher than 1 kV. That is, in the present embodiment, the breakdown voltage of the second limiting diode **231** is set so as not to fall below a differential voltage of a voltage value applied to the power supply line **150** with respect to a minimum value of a voltage applied to the other end **821** of the secondary coil **L2** during flow of a primary current through the primary coil **L1**. In other words, the breakdown voltage of the second limiting diode **231** is set so as not to fall below a differential voltage of a voltage value applied to the cathode side of the second limiting diode **231** with respect to a minimum value of a voltage applied to the anode side of the second limiting diode **231** in an ON state. Note that the minimum value of the voltage applied to the other end **821** of the secondary coil **L2** is, for example, minus 480 V. Meanwhile, the value of the voltage applied to the power supply line **150** is, for example, plus 12 V. Thus, no current flows through the second connecting wire **222a** in which the second limiting diode **231** is interposed.

(72) Further, the second resistor **232** is interposed in the second connecting wire **222b**. As described above, the resistance value of the second resistor **232** is equal to or higher than 10 MΩ. By setting the resistance value of the second resistor **232** to such a sufficiently high value as described, it is possible to sufficiently reduce a current flowing via the second connecting wire **222b** in an ON state. As a result of this, in an ON state, an ON-state voltage generated in the secondary coil **L2** and a voltage (secondary voltage) generated at the one end **822** of the secondary coil **L2** can be reduced. This can suppress occurrence of discharge in the spark plug **113** in an ON state, that is, at an abnormal timing.

(73) Further, for discharge control, the state of the switching element **70** is changed from a closed state to an open state, and a primary current flowing from the power supply device **102** to the primary coil **L1** is interrupted. As a result, a high negative voltage ranging from minus several thousands of volts to minus several tens of thousands of volts is induced at the one end **822** of the secondary coil **L2**. This causes electrical breakdown at the gap **d** in the spark plug **113**. Then, there is generated a current directed toward the center electrode **161** of the spark plug **113** from the

ground point (ground) **151** via the ground electrode **162** of the spark plug **113** (refer to FIG. 6). The thus generated current flows through the first connecting wire **221** and the secondary coil L2, and flows through the second limiting diode **231** in the forward direction or flows through the second resistor **232**. In the present embodiment, most of the current flows through the second limiting diode **231** in the forward direction while a part thereof flows through the second resistor **232**, and the current further flows to the ground point (ground) **153** via the power supply device **102**.

(74) Consequently, discharge occurs at the gap d in the spark plug **113**, to generate spark, so that a fuel supplied to the combustion chamber of the internal combustion engine is ignited. In addition, when the absolute value of the high negative value induced at the one end **822** of the secondary coil L2 falls below the discharge maintaining voltage at the gap d in the spark plug **113**, the discharge at the gap d in the spark plug **113** ends once.

(75) Further, in the same manner as in the first preferred embodiment, a diode having a breakdown voltage that is lower than the electrical breakdown voltage at the gap d in the spark plug **113** and the absolute value of the residual voltage value (De) at the one end **822** of the secondary coil L2 at the end of discharge is used as the second limiting diode **231**. The breakdown voltage of the second limiting diode **231** used in the present embodiment is equal to or lower than 2 kV. At that time, the residual voltage value (De) at the one end **822** of the secondary coil L2 is negative. Note that the residual voltage value (De) at the one end **822** of the secondary coil L2 is the residual voltage value (De) on the anode side of the second limiting diode **231**, and is, for example, minus several tens of thousands of volts. Meanwhile, the voltage applied to the power supply line **150** is, for example, 12 V. In other words, the voltage applied to the cathode side of the second limiting diode **231** is, for example, 12 V.

(76) This allows a current to flow suddenly and abundantly through the second limiting diode **231** in the reverse direction from the side where the power supply device **102** is present, without occurrence of further discharge in the spark plug **113** after the end of discharge. That is, a current (secondary current) flows toward the secondary coil L2 from the side where the power supply device **102** is present, via the second connecting wire **222a**.

(77) Consequently, electric charge remaining in spots such as a spot near the one end **822** of the secondary coil L2, the first connecting wire **221**, or a spot near the center electrode **161** of the spark plug **113** is removed, and thus the absolute value of the voltage (secondary voltage) applied to the one end **822** of the secondary coil L2 is significantly reduced at once, so that residual energy remaining in those spots can be reduced. Thus, the absolute value of the voltage (secondary voltage) applied to the one end **822** of the secondary coil L2 is significantly reduced to the substantially same value as the breakdown voltage of the second limiting diode **231** at once.

(78) Further, as described above, the second connecting wire **222b** is laid in parallel with the second connecting wire **222a** in which the second limiting diode **231** is interposed, and the second resistor **232** is interposed in the second connecting wire **222b**. After the voltage value at the one end **822** of the secondary coil L2 is reduced to the substantially same value as the breakdown voltage of the second limiting diode **231**, a current flows via the second connecting wire **222b**. More specifically, a current (secondary current) flows toward the secondary coil L2 from the side where the power supply device **102** is present, via the second connecting wire **222b** in which the second resistor **232** is interposed.

(79) Moreover, in the present embodiment, the resistance value of the second resistor **232** is equal to or lower than 50 MΩ. By setting the resistance value of the second resistor **232** to such a low value as described, it is possible to keep a certain amount of current or more flowing from the side where the power supply device **102** is present, via the second connecting wire **222b**, after the end of discharge. This enables further reduction of residual energy remaining near the one end **822** of the secondary coil L2, in the first connecting wire **221**, near the center electrode **161** of the spark plug **113**, or the like, and allows the energy to converge toward zero earlier.

(80) As is made clear from the foregoing description, in the present embodiment, first, for charge

control, when a primary current flows through the primary coil **L1** (in an ON state), a current flowing through the secondary coil **L2** can be suppressed by the second limiting diode **231** and the second resistor **232** connected in parallel. This enables reduction of an ON-state voltage generated in the secondary coil **L2**. Consequently, occurrence of discharge in the spark plug **113** in an ON state can be suppressed.

(81) Further, after the end of discharge, a current (secondary current) flows suddenly and abundantly through the second limiting diode **231** in the reverse direction from the side where the power supply device **102** is present, via the second connecting wire **222a**. Thus, electric charge remaining in spots such as a spot near the one end **822** of the secondary coil **L2**, the first connecting wire **221**, or a spot near the center electrode **161** of the spark plug **113** can be removed at once. As a result, an absolute value of a voltage value caused by residual energy remaining in those spots can be significantly reduced to the substantially same value as the breakdown voltage of the second limiting diode **231** at once, first. Further, after that, a current flows via the second resistor **232**, so that the absolute value of the voltage value caused by the residual energy remaining in those spots can be reduced toward zero earlier.

(82) Consequently, even in a case in which there is a change in the pressure in the internal combustion engine afterward, occurrence of discharge at the gap **d** in the spark plug **113** at an abnormal timing can be suppressed. This prevents a fuel from being ignited at an abnormal timing also in an internal combustion engine that uses a fuel including hydrogen having properties of high combustibility at relatively low temperatures and of a high combustion rate, leading to suppression of breakage of the engine and the like.

(83) Meanwhile, in the first preferred embodiment, the first limiting diode **131** and the first resistor **132** are provided on the side where the one end **822** of the secondary coil **L2** is present, as described above. In contrast thereto, in the present embodiment, the second limiting diode **231** and the second resistor **232** are provided on the side where the other end **821** of the secondary coil **L2** is present. In this regard, in many cases, a floating capacitor comprising a small amount of electrostatic capacitance component is present also in the secondary coil **L2** itself. For this reason, in a case in which the second limiting diode **231** and the second resistor **232** are provided on the side where the other end **821** of the secondary coil **L2** is present, an electrostatic capacitance component included in the secondary coil **L2** itself is superimposed on the above-described electrostatic capacitance component of approximately 15 to 20 pF formed between the one end **822** of the secondary coil **L2** and the spark plug **113**.

(84) Thus, the superimposed electrostatic capacitance components have greater influence, to cause a fear that the above-described ON-state voltage may be further increased and the residual energy remaining at the time when the above-described discharge ends once may be increased. In view of this, it is preferred that the limiting diode and the resistor be provided on the side where the one end **822** of the secondary coil **L2** is present. However, in terms of ease of securing a mounting space for the second limiting diode **231** and the second resistor **232**, it is preferred that the limiting diode and the resistor be provided on the side where the other end **821** of the secondary coil **L2** is present, as in the second preferred embodiment, in some cases.

3. Modifications

(85) The illustrative preferred embodiments of the present invention have been described above, but the present invention is not limited to the above-described preferred embodiments.

(86) According to the above-described embodiments and modifications, in charge control, a voltage applied to the one end **822** of the secondary coil **L2** is positive and a voltage applied to the other end **821** of the secondary coil **L2** is negative. Further, in discharge control, a high negative voltage ranging from minus several thousands of volts to minus several tens of thousands of volts is induced at the one end **822** of the secondary coil **L2**. Alternatively, the positive and negative of the voltage values at both ends **821** and **822** of the secondary coil **L2** may be reversed by a change in a direction of winding of the primary conductor **81** in the primary coil **L1** or a change in a direction

of winding of the secondary conductor **82** in the secondary coil **L2**. In this case, the forward direction and the reverse direction of the first limiting diode **131** interposed in the first connecting wire **121a** in the first preferred embodiment and the second limiting diode **231** interposed in the second connecting wire **222a** in the second preferred embodiment are reversed.

(87) According to the above-described first preferred embodiment, the cathode side of the first limiting diode **131** and the other end **821** of the secondary coil **L2** are connected to the positive side of the power supply device **102**. Alternatively, as shown in FIG. 7 that shows a first modification, the cathode side of the first limiting diode **131** and the other end **821** of the secondary coil **L2** may be connected to a ground point (ground) **154**. Meanwhile, according to the second preferred embodiment, the cathode side of the second limiting diode **231** and the other end **821** of the secondary coil **L2** are connected to the positive side of the power supply device **102**. Alternatively, as shown in FIG. 8 that shows a second modification, the cathode side of the second limiting diode **231** and the other end **821** of the secondary coil **L2** may be connected to the ground point (ground) **154**.

(88) Specifically, the second limiting diode **231** may be a Zener diode or an avalanche diode that is interposed in one of the two second connecting wires **222a** and **222b** laid in parallel between the other end **821** of the secondary coil **L2** and the ground point (ground) **154** and is forward-biased when it is oriented in a direction from the one end **822** to the other end **821** of the secondary coil **L2**. Further, the second resistor **232** may be interposed in the other of the two second connecting wires **222a** and **222b**.

(89) In the first modification and the second modification, first, for charge control, when a primary current flows through the primary coil **L1** (in an ON state), an ON-state voltage is generated across both ends **821** and **822** of the secondary coil **L2**. For example, a potential difference of 960 V is generated across the one end **822** and the other end **821** of the secondary coil **L2**. A maximum value of a voltage applied to the one end **822** of the secondary coil **L2** is positive, and a minimum value of a voltage applied to the other end **821** of the secondary coil **L2** is negative. For example, a maximum value of a voltage applied to the one end **822** of the secondary coil **L2** is approximately plus 480 V, and a minimum value of a voltage applied to the other end **821** of the secondary coil **L2** is approximately minus 480 V.

(90) It should be noted here that, in the first modification, the first limiting diode **131** and the first resistor **132** are interposed in the first connecting wires **121a** and **121b**. Meanwhile, in the second modification, the second limiting diode **231** and the second resistor **232** are interposed in the second connecting wires **222a** and **222b**. Each of the first limiting diode **131** and the second limiting diode **231** is forward-biased when it is oriented in a direction from the one end **822** to the other end **821** of the secondary coil **L2**. Thus, by flow of a current from the one end **822** to the other end **821** of the secondary coil **L2**, and further to the ground point (ground) **154**, an ON-state voltage or a secondary voltage generated in the secondary coil **L2** can be reduced. This can suppress occurrence of discharge in the spark plug **113** in an ON state, that is, at an abnormal timing.

(91) Further, for discharge control, the state of the switching element **70** is changed from a closed state to an open state, and a primary current flowing from the power supply device **102** to the primary coil **L1** is interrupted. As a result, a high negative voltage ranging from minus several thousands of volts to minus several tens of thousands of volts is induced at the one end **822** of the secondary coil **L2**. This causes electrical breakdown at the gap **d** in the spark plug **113**. Then, in the first modification, there is generated a current (secondary current) directed toward the center electrode **161** of the spark plug **113** from the ground point (ground) **151** via the ground electrode **162** of the spark plug **113** (refer to FIG. 7). The thus generated current flows through the first connecting wire **121a** in which the first limiting diode **131** is interposed, in the forward direction, or flows through the first connecting wire **121b** in which the first resistor **132** is interposed, flows from the one end **822** to the other end **821** of the secondary coil **L2**, and further flows toward the

ground point (ground) **154**.

(92) Meanwhile, in the second modification, there is generated a current (secondary current) that flows from the ground point (ground) **151** toward the center electrode **161** of the spark plug **113** via the ground electrode **162** of the spark plug **113** (refer to FIG. **8**). The thus generated current further flows from the one end **822** to the other end **821** of the secondary coil L2, and flows through the second connecting wire **222a** in which the second limiting diode **231** is interposed, in the forward direction, or flows through the second connecting wire **222b** in which the second resistor **232** is interposed. Further, the current flows toward the ground point (ground) **154**. Consequently, discharge occurs at the gap d in the spark plug **113**, to generate spark, so that a fuel supplied to the internal combustion chamber is ignited.

(93) Moreover, in the first modification, after the end of discharge, a current (secondary current) suddenly and abundantly flows through the first limiting diode **131** in the reverse direction from the ground point (ground) **154** via the secondary coil L2. As a result, electric charge remaining in spots such as a spot near the one end **822** of the secondary coil L2, the first connecting wire **121**, or a spot near the center electrode **161** of the spark plug **113** can be removed at once. Consequently, an absolute value of a voltage value caused by residual energy remaining in those spots can be significantly reduced to the breakdown voltage of the first limiting diode **131** at once, first. Further, after that, a current flows from the ground point (ground) **154** via the first resistor **132**, so that the absolute value of the voltage value caused by the residual energy remaining in those spots can be reduced toward zero earlier.

(94) Meanwhile, in the second modification, after the end of discharge, a current (secondary current) suddenly and abundantly flows through the second limiting diode **231** in the reverse direction from the ground point (ground) **154** toward the secondary coil L2. As a result, electric charge remaining in spots such as a spot near the one end **822** of the secondary coil L2, the first connecting wire **221**, or a spot near the center electrode **161** of the spark plug **113** can be removed at once. Consequently, an absolute value of a voltage value caused by residual energy remaining in those spots can be significantly reduced to the breakdown voltage of the second limiting diode **231** at once, first. Further, after that, a current flows from the ground point (ground) **154** via the second resistor **232**, so that the absolute value of the voltage value caused by the residual energy remaining in those spots can be reduced toward zero earlier.

(95) The ignition device of the present invention may be any device that is installed in various devices and industrial machines such as a power generator, in addition to a vehicle such as an automobile, and is used to generate electric spark in a spark plug of an internal combustion engine and ignite a fuel.

(96) The details of the shapes and configurations of the above-described ignition device may be appropriately changed within a range not departing from the gist of the present invention. Further, the respective elements described in the above-described preferred embodiments and modifications may be appropriately combined unless contradiction arises.

Claims

1. An ignition device for use in an internal combustion engine that uses a fuel including at least hydrogen, comprising: an ignition coil formed by electromagnetic coupling of a primary coil and a secondary coil; a power supply device configured to apply a direct-current voltage to one end of the primary coil via a power supply line; a switching element interposed between the other end of the primary coil and a ground point, the switching element being capable of performing switching between passage and interruption of a primary current flowing through the primary coil from the power supply device; a spark plug configured to ignite the fuel by occurrence of discharge at a gap, in accordance with a high voltage induced at one end of the secondary coil; a first limiting diode comprising a Zener diode or an avalanche diode that is interposed in one of two first connecting

wires laid in parallel between the one end of the secondary coil and the spark plug, and is forward-biased when the diode is oriented in a direction from the one end to the other end of the secondary coil; and a first resistor interposed in the other of the two first connecting wires, wherein the first limiting diode has a breakdown voltage that is equal to or higher than a value calculated by multiplication of a voltage value of the direct-current voltage applied to the one end of the primary coil by the power supply device by a ratio of the number of turns of the secondary coil to the number of turns of the primary coil, and is lower than a discharge maintaining voltage at the gap in the spark plug, and the first resistor has a resistance value that is equal to or higher than 10 MΩ and is equal to or lower than 50 MΩ.

2. The ignition device according to claim 1, wherein the breakdown voltage is equal to or higher than 1 kV.

3. The ignition device according to claim 1, wherein the breakdown voltage is equal to or lower than 2 kV.

4. The ignition device according to claim 1, further comprising a control unit configured to control the switching of the switching element, wherein the control unit performs charge control in which the switching element is placed in a closed state, so that a primary current flows through the primary coil, to charge the primary coil, and performs discharge control in which, after the charge control, a state of the switching element is changed to an open state and a high voltage is induced at the one end of the secondary coil, so that discharge occurs at the gap in the spark plug, and an absolute value of the voltage induced at the one end of the secondary coil at an ending point of the discharge control is higher than an absolute value of the voltage induced at the one end of the secondary coil at a starting point of the discharge control.

5. The ignition device according to claim 1, further comprising a floating capacitor formed between the one end of the secondary coil and the spark plug.

6. An ignition device for use in an internal combustion engine that uses a fuel including at least hydrogen, comprising: an ignition coil formed by electromagnetic coupling of a primary coil and a secondary coil; a power supply device configured to apply a direct-current voltage to one end of the primary coil via a power supply line; a switching element interposed between the other end of the primary coil and a ground point, the switching element being capable of performing switching between passage and interruption of a primary current flowing through the primary coil from the power supply device; a spark plug configured to ignite the fuel by occurrence of discharge at a gap, in accordance with a high voltage induced at one end of the secondary coil; a second limiting diode comprising a Zener diode or an avalanche diode that is interposed in one of two second connecting wires laid in parallel between the other end of the secondary coil and the power supply device or the ground point, and is forward-biased when the diode is oriented in a direction from the one end to the other end of the secondary coil; and a second resistor interposed in the other of the two second connecting wires, wherein the second limiting diode has a breakdown voltage that is equal to or higher than a value calculated by multiplication of a voltage value of the direct-current voltage applied to the one end of the primary coil by the power supply device by a ratio of the number of turns of the secondary coil to the number of turns of the primary coil, and is lower than a discharge maintaining voltage at the gap in the spark plug, and the second resistor has a resistance value that is equal to or higher than 10 MΩ and is equal to or lower than 50 MΩ.

7. The ignition device according to claim 6, wherein the breakdown voltage is equal to or higher than 1 kV.

8. The ignition device according to claim 6, wherein the breakdown voltage is equal to or lower than 2 kV.
